CSE205

Object Oriented Programming Language

COURSE CODE: CSE 205
COURSE TITLE: OBJECT ORIENTED PROGRAMMING

Courtesy: Lec Sunjida Nasrin Tumpa

Generic Programming

&

Templates

An old example (from C)

```
int abs (int n){
  return (n < 0) ? -n : n;
}
long labs (long n){
  return (n < 0) ? -n : n;
}
float fabs (float n){
  return (n < 0) ? -n : n;
}
```

```
main(){
  int i=-3;
  cout<<abs (i);
  long l=-7;
  cout<< labs (l);
  float f=-2.0;
  cout<<fabs (f);
 }
```

An old example (C++ : Function Overloading)

```
int abs (int n){
  return (n < 0) ? -n : n;
}
long abs (long n){
  return (n < 0) ? -n : n;
}
float abs (float n){
  return (n < 0) ? -n : n;
}
```

```
main(){
  int i=-3;
  cout<<abs (i);
  long l=-7;
  cout<< abs (l);
  float f=-2.0;
  cout<<abs (f);
 }
```

Example (C++ : Template)

Generic/ Template Function

```
template <class T>
T abs(T n){
    return (n < 0) ? -n : n;
}
```

```
main(){
    int i=-3;
    cout<<abs (i);
    long l=-7;
    cout<< abs (l);
    float f=-2.0;
    cout<<abs (f);
}
```

Generic Function

- Programming that works regardless of type is called generic programming.
- A generic function defines a general set of operations that will be applied to various types of data.
- Another type of polymorphism, known as parametric polymorphism.
- Give the user the ability to reuse code in a simple, type-safe manner that allows the compiler to automate the process of type *instantiation*.

Function Templates

- A generic function is created using the keyword template.
- Also called Function Template

Declaration Syntax :

```
template <class identifier>

return_type function_name (arguments of type identifier) { … }
```

```
template <typename identifier>

return_type function_name (arguments of type identifier) { … }
```

- Function return type does not take into account to select the template function (just like function overloaded).

What the compiler does ?

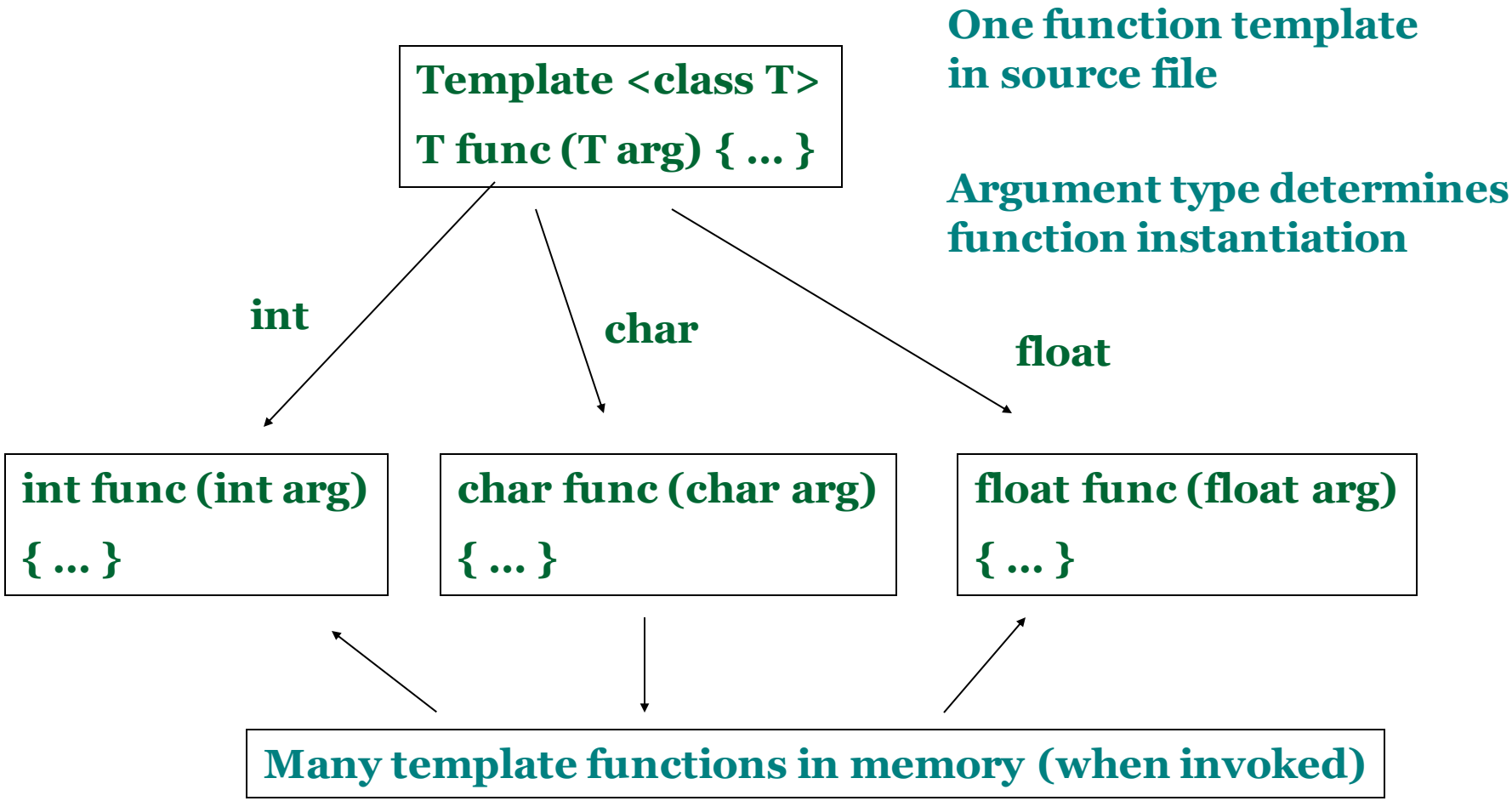
- Function template does not cause the compiler to generate any code (similar to the way a class does not create anything in memory).

- Compiler simply remembers the template for possible future use.

- Code generation does not take place until the function is actually called (invoked).

- Compiler generate a template function (specific version of a function template) depends on the function signature (that's why we called parametric polymorphism).

A Function Template



Function templates with multiple arguments

```
// two arguments template
template<class T1, class T2>
void copy (T1 a[], T2 b[], int n){
   for (int i = 0; i < n; ++i)
         a[i] = b[i];
}
```

```
main(){
  char c[50] = {‘a’, ‘b’, ‘d’, ‘e’, ‘f’};
  int i[50] = {10, 20 30, 40, 50};
  float f[50] = {1.1, 2.2, 3.3, 4.4, 5.5};
  copy (i, f, 5);   // ok, but loss of data
  copy (c, f, 5);  // ok, but loss of data
  copy (f, i, 5);   // ok, int to float
  copy (f, c, 5);  // ok, char to float
}
```

Function templates with multiple arguments

```
template <typename T> void swap( T& x, T& y){
    T tmp;
    tmp = x;
    x = y;
    y = tmp;
}
```

```
main(){
        int x, y;
        char c;
        double m, n;
        swap( x, y );
        swap( m, n );
        swap( x, m ); // error
}
```

Generic Function Vs Function Overloading

- When functions are overloaded, one can have different actions performed within the body of each function but generic function must perform the same general version for all actions.

- A generic function can be overloaded. In that case the overloaded function overrides the generic function relative to that specific version.

Generic Function Vs Function Overloading

```
template <class X>
void func(X n){
    cout << n << endl;
}
```

```
template <class X, class Y>
void func(X a, Y b){
    cout <<a << b << endl;
}
```

```
main(){
    func(10);
    func(20, 30);
}
```

Generic Function Vs Function Overloading

```
template <class T>
T func(T n){
    cout<<"Inside Generic Function: "<<n;
}
```

```
int func(int n){
    cout<<"Inside Specific Function: "<<n;
}
```

```
main(){
  int i=3;
  cout<< func(i);
  double d=7;
  cout<< func (d);
  float f=-2.0;
  cout<< func (f);
}
```

Generic Class

- A generic class defines the algorithm used by the class but the actual type of data is specified when object of that class are created.

```
template < class identifier >
class classname { ... };
```

```
classname<type> ob; //Object Instantiation
```

- The *identifier* is a template argument that essentially stands for an arbitrary type.
- The template argument can be used as a type name throughout the class definition.

Template class stack

```
// template stack implementation
template <class TYPE>
class stack {
  TYPE*  s;
  int  top;
  public:
    stack (int size = 100){
          s = new TYPE[size];
          top=0;
    }
   ~stack() { delete []s; }
    void  push (TYPE c){
            s[top++] = c;   }
    TYPE  pop() {
            return s[--top];  }
};
```

```
void main(){

 // 100 char stack
 stack<char>    stk_ch;

 // 200 int stack
 stack<int>   stk_int(200);

 // 20 float stack
 stack<float>   stk_float(20);
}
```